

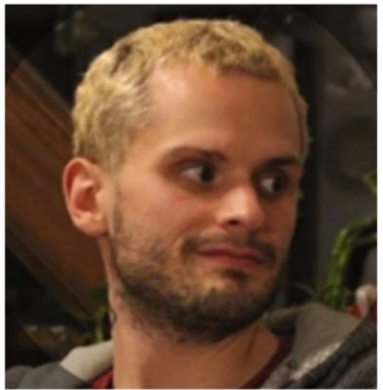


**wellcome
centre
integrative
neuroimaging**



INTRODUCTION TO OPEN SCIENCE – PART 2

How to share open science data
@ WIN



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First: a poll!

- Were you there for the first Introduction to Open Science?



Open science concepts

- Increasing the openness, integrity, and reproducibility of research.
- Transparency in methodology, data collection and analysis.
- Availability and reusability of scientific data.
- Accessibility and transparency of scientific communication.
- Facilitating scientific collaboration

Estimating the reproducibility of psychological science

Open Science Collaboration^{*,†}

+ See all authors and affiliations

Science 28 Aug 2015;
Vol. 349, Issue 6251, aac4716
DOI: 10.1126/science.aac4716

Article

Figures & Data

Info & Metrics

eLetters

 PDF

Empirically analyzing empirical evidence

One of the central goals in any scientific endeavor is to understand causality. Experiments that seek to demonstrate a cause/effect relation most often manipulate the postulated causal factor. Aarts *et al.* describe the replication of 100 experiments reported in papers published in 2008 in three high-ranking psychology journals. Assessing whether the replication and the original experiment yielded the same result according to several criteria, they find that about one-third to one-half of the original findings were also observed in the replication study.

Science, this issue [10.1126/science.aac4716](https://doi.org/10.1126/science.aac4716)

Reasons to
share your
data

Concerns over loss of
public confidence in
science e.g.
'reproducibility crises'

		Data	
		Same	Different
Analysis/ Code	Same	Reproducible	Replicable
	Different	Robust	Generalisable

adapted from Kirstie Whitaker

Reasons to
share your
data

Make your experiment
and results reproducible

Policy on data, software and materials management and sharing

As a charity, Wellcome works to ensure that the results of the research we fund are applied for the public good. This includes creating an environment that enables and incentivises researchers to maximise the value of their research outputs, including data, software and materials.

Policy statement

1. We expect our researchers to maximise the availability of research data, software and materials with as few restrictions as possible. As a minimum, the data underpinning research papers should be made available to other researchers at the time of publication, as well as any original software that is required to view datasets or to replicate analyses. Where research data relates to public health emergencies, researchers must share quality-assured interim and final data as rapidly and widely as possible, and in advance of journal publication.

Wellcome website

Reasons to share your data

Increasing demands of funders to share raw data

Common principles on data policy

The Research Council common principles on data policy provide an overarching framework for individual Research Council policies on data policy.

The common principles are listed below and further guidance can be found within the [guidance documentation \(PDF, 490KB\)](#).

Information on supporting research data management costs through grant funding can be found [here](#).

Research Council common principles on data

- Publicly funded research data are a public good, produced in the public interest, which should be made openly available with as few restrictions as possible in a timely and responsible manner.
- Institutional and project specific data management policies and plans should be in accordance with relevant standards and community best practice. Data with acknowledged long-term value should be preserved and remain accessible and usable for future research.
- To enable research data to be discoverable and effectively re-used by others, sufficient metadata should be recorded and made openly available to enable other researchers to understand the research and re-use potential of the data. Published results should always include information on how to access the supporting data.

UKRI website

Reasons to
share your
data

Increasing demands of
funders to share raw
data

As the research community embraces data sharing, academic journals can do their bit to help.

Starting this month, all research papers accepted for publication in *Nature* and an initial 12 other Nature titles will be required to include information on whether and how others can access the underlying data.

These statements will report the availability of the 'minimal data set' necessary to interpret, replicate and build on the findings reported in the paper. Where applicable, they will include details about publicly archived data sets that have been analysed or generated during the study. Where restrictions on access are in place — for example, in the case of privacy limitations or third-party control — authors will be expected to make this clear.

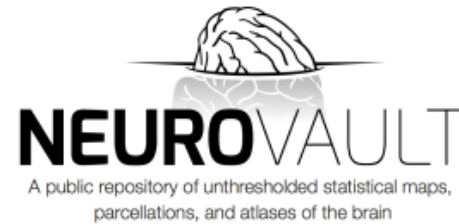
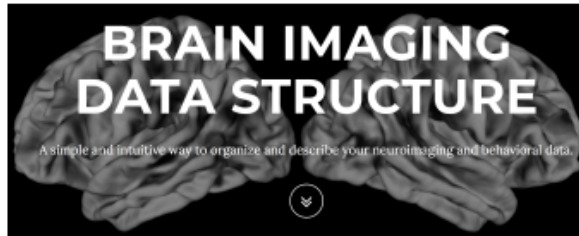
Related stories

- [The ups and downs of data sharing in science](#)
- [Data sharing: Access all areas](#)
- [Data-access practices](#)

Nature, 2016

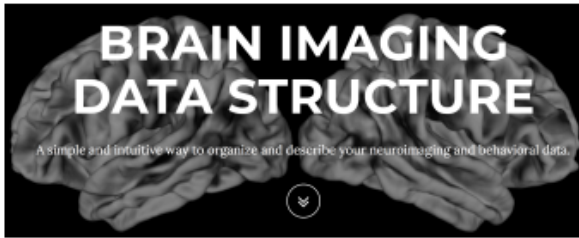
Reasons to share your data

Major journals mandate data sharing as part of publication process



Reasons to
share your
data

Development of
(better, and more
stable) online
repositories and formats
for data sharing



GitHub



Our subject for
today!

Reasons to
share your
data

development of
, and more
online

repositories and formats
for data sharing

What to share?

- (MR) Protocols
- Tasks
- (MRI) Data
- Analysis code

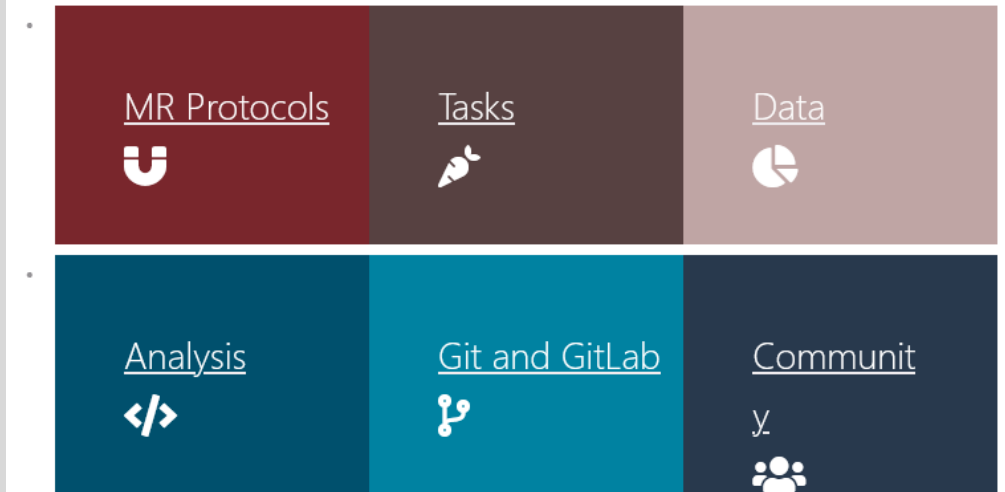
Open Research at the Wellcome Centre for Integrative Neuroimaging (WIN)

Find out how to share your research outputs effectively and responsibly, for improved impact, access and collaboration

This repository is the hub for information on how to employ open research practices at the [Wellcome Centre for Integrative Neuroimaging \(WIN\)](#).

The material shared here is written and maintained by an active community of WIN researchers, and is intended to support all WIN members in using the sharing infrastructure created by the [WIN Open Neuroimaging Project](#).

Jump straight into one of the sections below, or use the sidebar to find out more about this repository.



How to share?

- Always check all the items before starting a study

<https://open.win.ox.ac.uk/pages/open-science/community/Open-WIN-Community/docs/data/can-i/>



= Check with your supervisor/lab members/department/University



⚠ Do you own the IP?	Did you collect the data? If secondary analysis, check the license or usage agreement
⚠ Industry partners	May have imposed conditions on sharing to protect their commercial interests. Review your contract for details
⚠ Commercial potential	Speak to Oxford University Innovation
⚠ DP[I]A	Your assessment of the risk and mitigations of data breach. What sharing activities were described?
✅ Data Management Plan	What sharing did you plan for?
⚠ Is GDPR relevant?	Is your data from living humans? If preclinical or ex vivo, skip to “metadata”

IP = Intellectual property

DPA = Data processing agreement

an agreement between a data controller (such as a company) and a data processor (such as a third-party service provider).



⚠ Participant Info Sheet	What did you describe? Use CUREC AP17 where possible
⚠ Consent	Did your participant consent?

State in your ethics that you will publish data in Open Science!!!



Scans, recordings or other non-invasive tests (adults)

+ Expand All

AP03 Electrophysiological (EEG) recordings from the scalp



AP08 Magnetoencephalographic (MEG) recordings from adult volunteers



AP17 Non-invasive magnetic resonance investigations (MRI)



➤ AP17 CUREC approved procedure IDREC

➤ AP17 Consent form

➤ AP17 Participant information sheet

➤ AP17 3T Volunteer MRI safety screening form

This PDF is not accessible because it is provided for printing only.

➤ AP17 7T Volunteer MRI safety screening form

This PDF is not accessible because it is provided for printing only.

If applicable: I / We may use data from this research in future studies and **share** this with other researchers (e.g. in online databases). This will only be in a form that does not identify you. [\[Refer to ICO guidance on anonymisation to achieve necessary standard. See in particular appendix 2 and annex 1.\]](#)

Procedure 17 and its associated documents above are for non-invasive magnetic resonance investigations in research participants.



⚠ Direct identifiers removed	e.g. Name, address, telephone number, NHS number
⚠ Indirect identifiers aggregated	e.g. age, gender, handedness. Bin so no participant is unique ($n \geq 5$)
⚠ Keys protected	Mapping between pxID and 1) direct identifiers 2) WIN scan ID. Stored only on University systems and separate from the data
⚠ Defaced	With quality assessment / confirmation
⚠ Participant space	Brain is a fingerprint
⚠ Unique dicom and .json fields	Consider [cumulative] risk of identification of retaining unique fields

pxID = Partner Cross Identifier

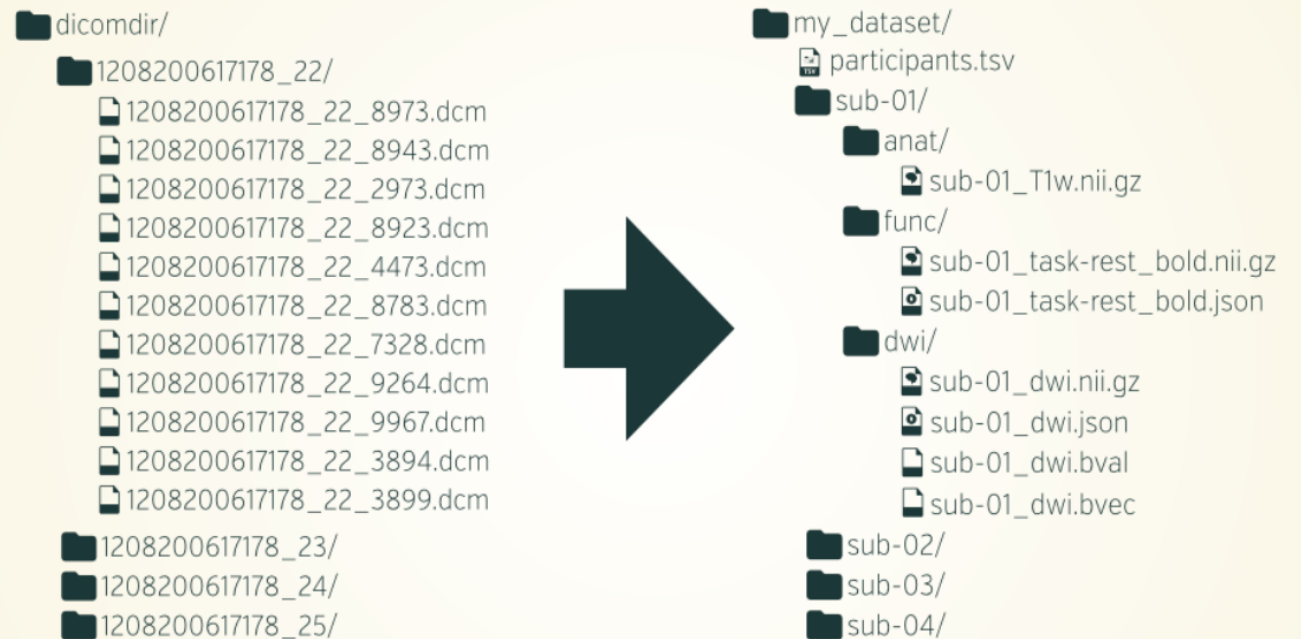


✓ Is it FAIR	DOIs, available, non-proprietary formats, licensed
✓ Quality control analysis	Share outcome
✓ Acquisition methodology	MR Protocols data base, or other tools
✓ Data dictionary	Single machine readable file containing all covariates
✓ Community standards	e.g. BIDS, Neurodata Without Borders
✓ Full experimental protocol	BIDS dataset_descriptor.json, or other record

Specifics about MRI data

- BIDS = Brain Imaging Data Structure
 - a simple and easy way of organizing neuroimaging and behavioral data
- BIDS is an established and evolving community standard for a multitude of neuroimaging data (MRI, (i)EEG, MEG, ...).
- It defines a data organization, naming schemes for files, and meta data descriptors.

bids.neuroimaging.io





✓ Reviewer only link	Before wider release
✓ Identifying bonafide researchers	Fully public (anonymous only) or managed access
✓ DOI	+/- versioning
✓ License	CC-BY-4.0 (data, docs) or Apache 2.0 (code)
✓ Authorship	Custom data usage agreement
✓ Reidentification	Custom data usage agreement
⚠ Funder requirements	e.g. acknowledgement in perpetuity

Where to share?

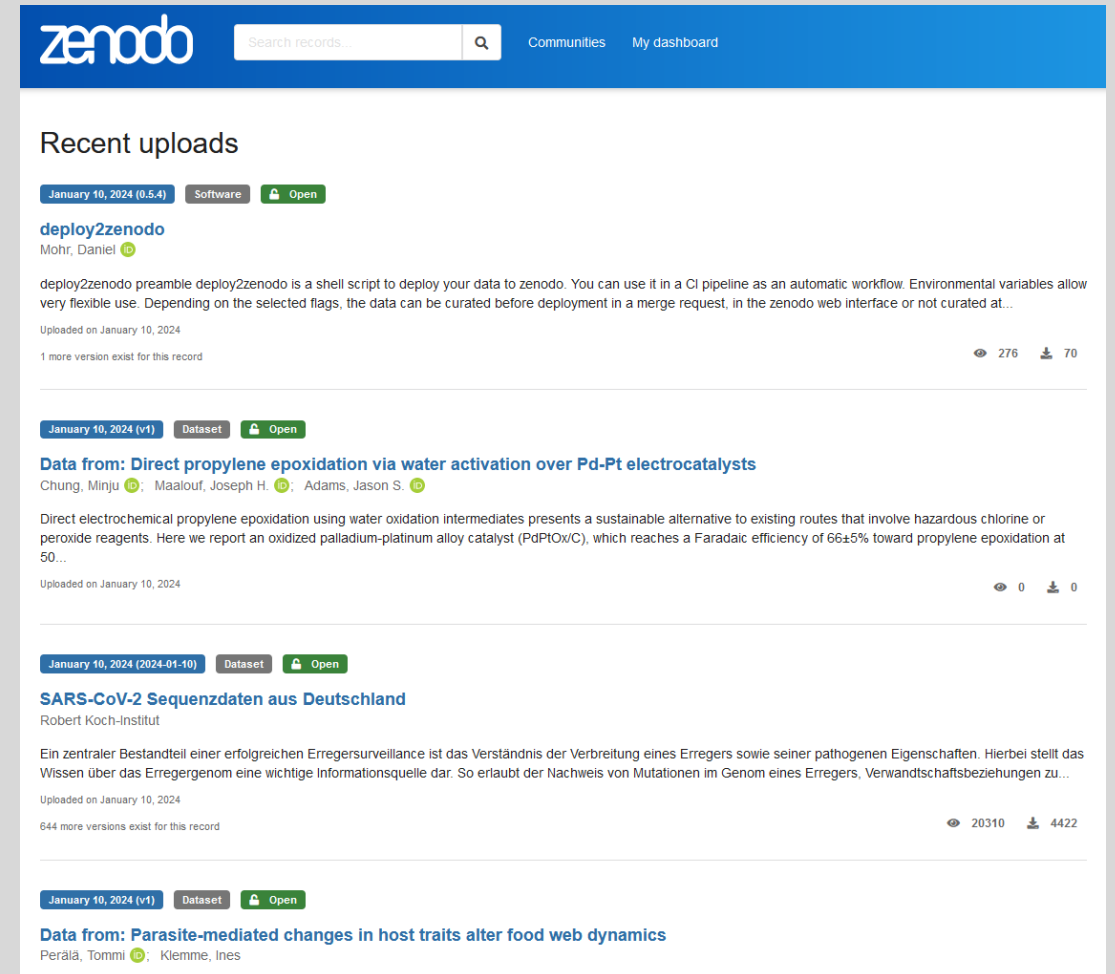
- OSF & Zenodo
 - All data formats (with size limits)
 - Anonymous data only
 - Integrates DOI (and versions for Zenodo)
 - Any CC license accepted
 - Allows embargo
 - Uses Reviewer only links
- Neurovault - A public repository of unthresholded statistical maps, parcellations, and atlases of the brain.
- Ebrains - An open research infrastructure that gathers data, tools and computing facilities for brain-related research
- OpenNeuro - A free and open platform for validating and sharing BIDS-compliant MRI, PET, MEG, EEG, and iEEG data
- WIN Open Data Portal – for WIN members
- WIN Digital Brain Bank - Neuroimaging data in post-mortem brains collected by University of Oxford researchers & collaborators



Things to keep in mind

↳ Citation

- Use Zenodo to make your shared data/code/elements citable
- Make your shared stuff citable
- Zenodo allows you to issue a new “version” of your record.
All versions will resolve to a single doi, therefore anyone who follows your old doi will see that a new version is available.



The screenshot displays the Zenodo homepage with a blue header containing the logo, a search bar, and links to 'Communities' and 'My dashboard'. The main content area is titled 'Recent uploads' and lists three records:

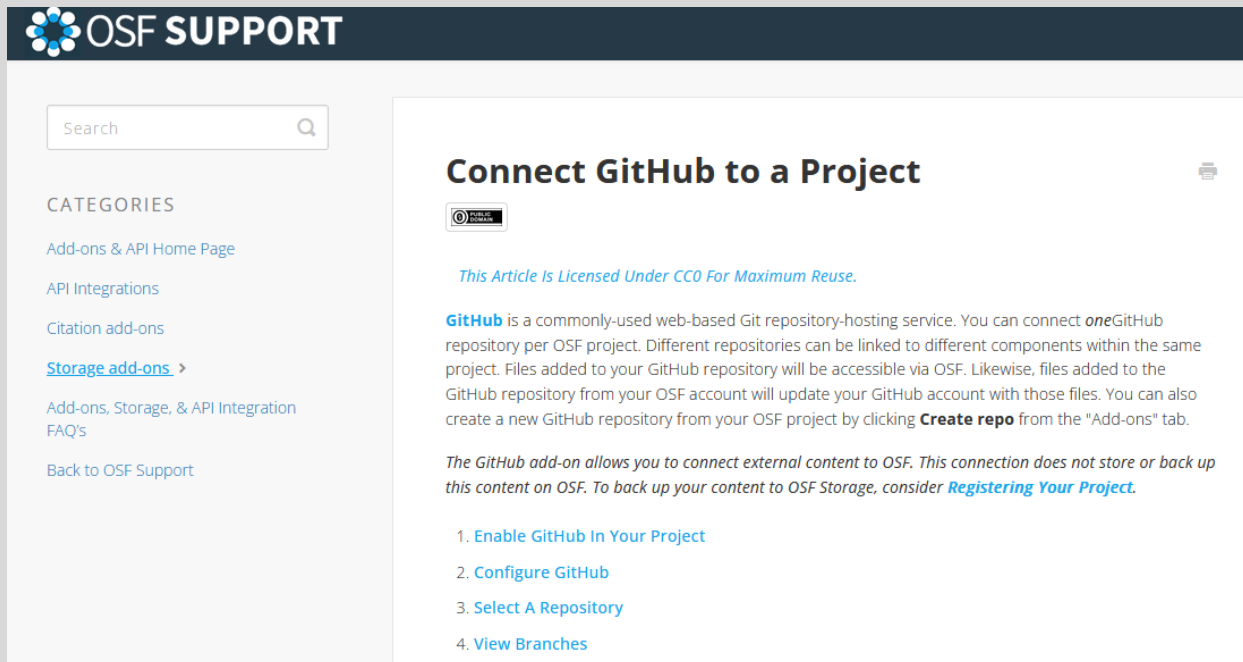
- deploy2zenodo** by Mohr, Daniel (ID). A software upload from January 10, 2024 (0.5.4). The description states it's a shell script for CI pipeline deployment. It has 276 views and 70 downloads.
- Data from: Direct propylene epoxidation via water activation over Pd-Pt electrocatalysts** by Chung, Minju (ID), Maalouf, Joseph H. (ID), and Adams, Jason S. (ID). A dataset upload from January 10, 2024 (v1). The description discusses a sustainable alternative to hazardous chlorine or peroxide reagents. It has 0 views and 0 downloads.
- SARS-CoV-2 Sequenzdaten aus Deutschland** by Robert Koch-Institut. A dataset upload from January 10, 2024 (2024-01-10). The description mentions its role in disease surveillance. It has 20310 views and 4422 downloads.

Each record includes a version label, category (Software or Dataset), an 'Open' button, a brief description, the upload date, and view/download statistics.

Things to keep in mind

↳ Tool integration

- OSF and Zenodo have Github integration (one repository per project)



The screenshot shows the OSF SUPPORT page. The header is dark blue with the OSF logo and 'OSF SUPPORT' text. Below the header is a search bar. On the left, there is a 'CATEGORIES' sidebar with links: 'Add-ons & API Home Page', 'API Integrations', 'Citation add-ons', 'Storage add-ons >', 'Add-ons, Storage, & API Integration FAQ's', and 'Back to OSF Support'. The main content area is titled 'Connect GitHub to a Project' and includes a Creative Commons license icon, a CC0 license notice, and a paragraph explaining how to connect a GitHub repository to an OSF project. It mentions that files added to the GitHub repository will be accessible via OSF and that a new GitHub repository can be created from the OSF project by clicking 'Create repo' from the 'Add-ons' tab. A note states that the GitHub add-on does not store or back up content on OSF. At the bottom, there is a numbered list of steps: 1. Enable GitHub In Your Project, 2. Configure GitHub, 3. Select A Repository, and 4. View Branches.

OSF SUPPORT

Search

CATEGORIES

- Add-ons & API Home Page
- API Integrations
- Citation add-ons
- [Storage add-ons >](#)
- Add-ons, Storage, & API Integration FAQ's
- Back to OSF Support

Connect GitHub to a Project

 This Article Is Licensed Under CC0 For Maximum Reuse.

GitHub is a commonly-used web-based Git repository-hosting service. You can connect *one* GitHub repository per OSF project. Different repositories can be linked to different components within the same project. Files added to your GitHub repository will be accessible via OSF. Likewise, files added to the GitHub repository from your OSF account will update your GitHub account with those files. You can also create a new GitHub repository from your OSF project by clicking **Create repo** from the "Add-ons" tab.

The GitHub add-on allows you to connect external content to OSF. This connection does not store or back up this content on OSF. To back up your content to OSF Storage, consider [Registering Your Project](#).

1. [Enable GitHub In Your Project](#)
2. [Configure GitHub](#)
3. [Select A Repository](#)
4. [View Branches](#)

Why use Zenodo?

- **Safe** — your research is stored safely for the future in CERN's Data Centre for as long as CERN exists.
- **Trusted** — built and operated by CERN and OpenAIRE to ensure that everyone can join in Open Science.
- **Citeable** — every upload is assigned a Digital Object Identifier (DOI), to make them citable and trackable.
- **No waiting time** — Uploads are made available online as soon as you hit publish, and your DOI is registered within seconds.
- **Open or closed** — Share e.g. anonymized clinical trial data with only medical professionals via our restricted access mode.
- **Versioning** — Easily update your dataset with our versioning feature.
- **GitHub integration** — Easily preserve your GitHub repository in Zenodo.
- **Usage statistics** — All uploads display standards compliant usage statistics

Things to keep in mind

↳ Metadata

- Do not forget the metadata!

Metadata is the data providing information about one or more aspects of the data; it is used to summarize basic information about data that can make tracking and working with specific data easier.

- In Github: metadata = readme file

The screenshot shows the GitHub repository page for 'List of Free Learning Resources In Many Languages'. At the top, there are links for README, Code of conduct, and CC-BY-4.0 license. The repository title is 'List of Free Learning Resources In Many Languages'. Below the title, there are badges for 'awesome', 'License CC BY 4.0', and 'Hacktoberfest 2023 is over! (1.3k PRs opened)'. A search bar is present with the text 'Search the list at <https://ebookfoundation.github.io/free-programming-books-search/>' and a 'Dynamic search site up' button. Below this, it says 'This page is available as an easy-to-read website. Access it by clicking on [Static site up](#).' The 'Intro' section explains that the list was originally a clone of 'StackOverflow - List of Freely Available Programming Books' and was moved to GitHub by Victor Felder. It also mentions that the 'Free Ebook Foundation' now administers the repo. The 'How To Contribute' section asks users to read the 'CONTRIBUTING' file and provides links to the 'Contributor Covenant 1.3', 'Code of Conduct', and 'translations'. At the bottom, there are several badges showing repository statistics: 'Issues 22 open', 'Good First issues 2 open', '"Help Wanted" issues 7 open', 'PRs 13 open', 'Merged PRs 6.2k', and '"Help Wanted" PRs 3 open'.

README Code of conduct CC-BY-4.0 license

List of Free Learning Resources In Many Languages

awesome License CC BY 4.0 Hacktoberfest 2023 is over! (1.3k PRs opened)

Search the list at <https://ebookfoundation.github.io/free-programming-books-search/> [Dynamic search site up](#).

This page is available as an easy-to-read website. Access it by clicking on [Static site up](#).

Intro

This list was originally a clone of [StackOverflow - List of Freely Available Programming Books](#) with contributions from Karan Bhangui and George Stocker.

The list was moved to GitHub by Victor Felder for collaborative updating and maintenance. It has grown to become one of [GitHub's most popular repositories](#).

Forks 59k Stars 307k Contributors 3.1k Sponsors 12 Watchers 9.6k Repo Size 18.7 MB

The [Free Ebook Foundation](#) now administers the repo, a not-for-profit organization devoted to promoting the creation, distribution, archiving, and sustainability of free ebooks. [Donations](#) to the Free Ebook Foundation are tax-deductible in the US.

How To Contribute

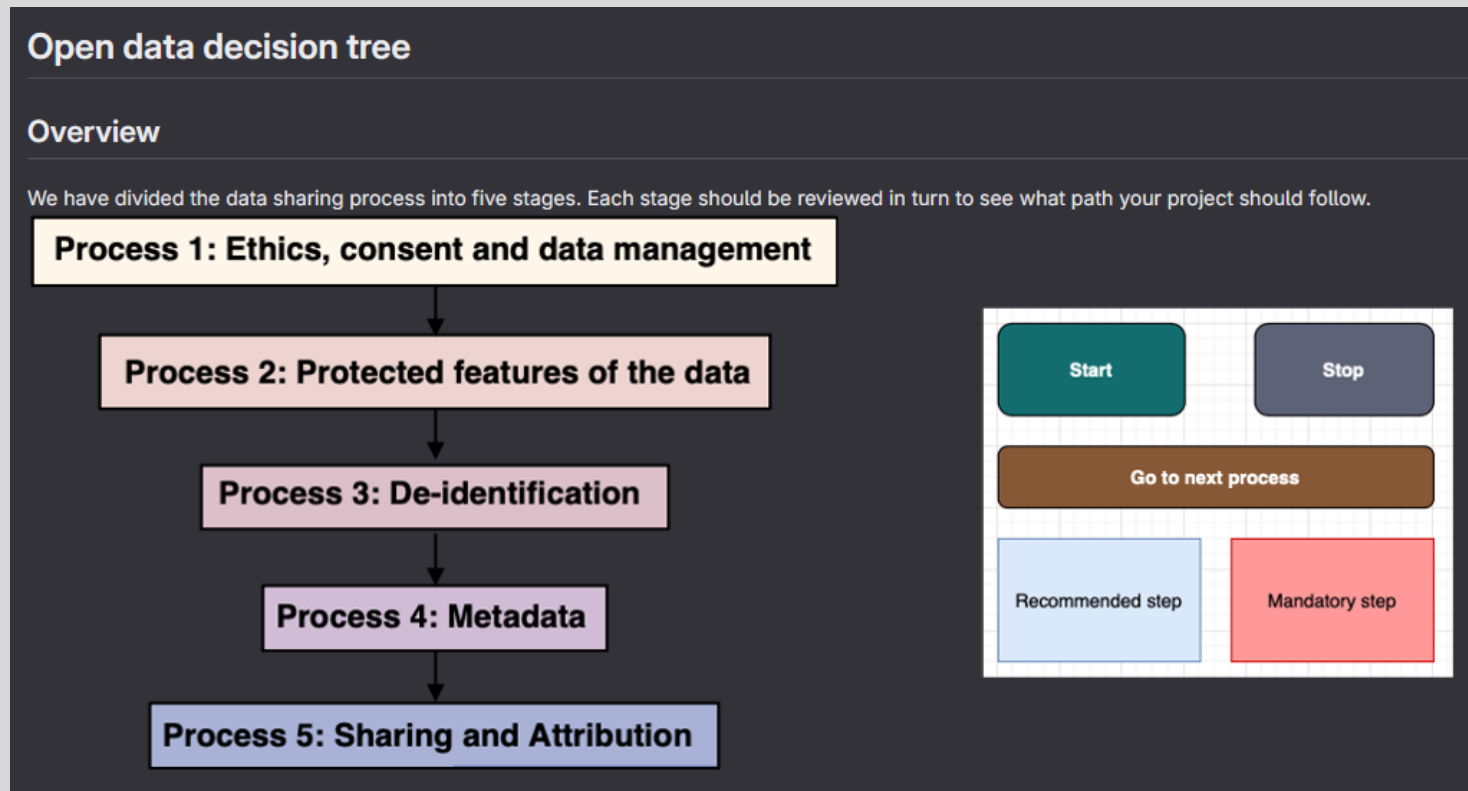
Please read [CONTRIBUTING](#). If you're new to GitHub, [welcome!](#) Remember to abide by our adapted from [Contributor Covenant 1.3](#) [Code of Conduct](#) too ([translations](#) also available).

Click on these badges to see how you might be able to help:

Issues 22 open Good First issues 2 open "Help Wanted" issues 7 open PRs 13 open Merged PRs 6.2k "Help Wanted" PRs 3 open

Things to keep in mind

↳ WIN community pages – learn more open science



<https://git.fmrib.ox.ac.uk/open-science/community/data-sharing-decision-tree/-/blob/master/docs/decision-tree.md>

Thank you!

If you have any questions
open@win.ox.ac.uk

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